Samuel Echefu

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Education

University of Victoria (UVIC) Sep 2024 - Apr 2028
Bachelor of Science in Computer Science (3rd year)

- **Completed Coursework:** Data Structures and Algorithms, Operating Systems, Software Systems Development, Machine Learning, Probability and Statistics, Calculus I-IV, Discrete Math

Skills

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL
Frameworks: React, React Native, Next.js, Express.js, Node.js, Flask, FastAPI, Tailwind, SpringBoot, Pandas
Databases & Cloud: MongoDB, PostgreSQL, Firebase, Supabase, Google Cloud Platform (GCP)
Other Tools: Git/Github, Figma, Docker, VSCode, STM32IDE, Expo SDK, JUnit, Jest

Work Experience

Software Developer | Aeternum Corp, Edmonton, AB May 2026 - Jul 2026

- Built context-processing logic that structures data into chunks and metadata for an LLM response pipeline
- Created a dashboard API that batched metric requests and reduced network overhead and cut latency by **70%**
- Designed and ported a Figma Prototype to a React Native mobile app to display data and LLM responses
- Implemented a permission-first RAG memory system using FastAPI and TypeScript, enforcing a **90%** confidence threshold and keeping user data safe and relevant per session

Software Engineer | UVic Satellite Design, Victoria, BC Sep 2026 - Present

- Bridged hardware components using a Raspberry Pi to relay SPI telemetry and video packets to a host PC
- Built a high-speed SPI video pipeline between an nRF5340, a Raspberry Pi, and an output laptop using FreeRTOS, Python, and C functions to chunk frames and stream in real time

Frontend Developer | ProperSeer, Calgary, AB May 2025 - Sep 2025

- Created responsive user interfaces using **React, Redux,** and **Bootstrap**, enhancing cross-device usability and caching expensive operations through memoization and prefetching, improving overall **page load time by 20%**
- Developed centralized dashboard views using React and Redux, streamlining property management workflows

Projects

MarketQuad | Next.js, React Native, Supabase, TypeScript, Cloudinary

- Launched a marketplace serving **20+** UVic students to buy & sell local items (market-quad.com)
- Improved API performance to **p95 < 500ms** at **1500+ req/s;** tested with **Locust** and **Next.js**
- Developed secure authentication with **Supabase** and **Postgres RLS**, restricting access to "@uvic.ca" emails
- Optimized image operations by utilizing **Cloudinary CDN** and prefetching to decrease listing load times
- Implemented a real-time messaging system using **Socket.io** for constant communication across devices

Benny | TypeScript, React Native, Java, Spring Boot, Google Cloud Platform

- Created a receipt-parsing backend in **Spring Boot** integrating the **Gemini API** to extract line items, structure and classify expense type, and common spending trends into **Firestore**
- Built a **React Native** mobile application paired with an automated data pipeline in **TypeScript** to visualize expense metrics with **Tailwind** and using weighted averages to predict grocery totals before upcoming shopping trips

uEyes - Google AI Hackathon | React, Tailwind, FastAPI, Firebase, Google Cloud Platform

- Designed a computer vision application for Google AI Studio's hackathon which utilizes the **Gemini API** to describe a user's environment with audio in **500ms** by maintaining short context windows for faster responses
- Developed using WebSockets for a real-time connection, **FastAPI, Firebase, Python, React**, and **Tailwind**
- Containerized the backend with **Docker** and deployed to **Google Cloud Platform (GCP)** for scalable hosting